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- GHG emissions are up to 93% higher in regions with a more carbon-intensive grid mix

The bigger picture

The growing demand for fast, contactless deliveries has been driving firms to experiment with automated package-delivery vehicles that can avoid urban traffic and rapidly reach customers. Widespread adoption of uncrewed aerial vehicles (UAVs) to replace a portion of first/last-mile truck pickups and deliveries could

reshape transportation energy by changing demand patterns and shifting fuel use from fossil fuels to electricity. There is a need to optimize drones and systems to maximize energy productivity. Here, we help stakeholders and researchers understand the energy use of package-delivery drones and provide an open model based on 188 delivery-drone flights with a range of payloads, speeds, and altitudes. We show the impact of the cruise speed and payload mass on the drone's range, provide generalizable coefficients for others to estimate drone energy use, and show that the energy per package delivered by drones can be up to 94% lower than conventional transportation modes.

Summary

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Keywords

quadcopter drone; last-mile delivery; energy consumption; greenhouse gas emissions; robot delivery; autonomous delivery

Data science maturity

DSML 2: Proof-of-concept: Data science output has been formulated, implemented, and tested for one domain/problem

Introduction

Achieving large improvements in the energy productivity of freight transportation is challenging, especially in the overwhelmingly petroleum-powered transport sector where medium and heavy trucks in the United States comprise 24% of transportation energy use. This sector is responsible for 37% of transportation-related greenhouse gas (GHG) emissions, while light-duty vehicles comprise 57% of transportation GHG emissions and 64% of transportation energy use. In addition, transportation remains a large source of nitrogen oxides (NO_x) and other air po

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speeds that vary from 4 to 12m/s in flights operating under wind conditions varying from 2 to 16 knots. We show that, within these limits, the ground speed and the wind condition impact little on the average energy consumption of the quadcopter drone and that the induced power at a hover and no-wind condition can be used as a good estimator of the average power experienced by the aircraft throughout the flight.

The adoption of a multirotor UAV was motivated by the commercial use of similar aircraft by last-mile delivery companies. For example, the company SpeedBird Aero has used a multirotor with a capacity of 2 kg to deliver food in Latin America.²⁸ The Irishdrone startup Manna²⁹ uses a quadcopter drone to carry payloads varying between 2 and 4 kg.³⁰ The DJI Matrice 100 (M100) used in this study is a smaller quadcopter drone and was tested with a maximum payload of 0.5 kg. The M100 carrying a very small payload is likely smaller than purpose-built package-delivery drones for larger packages, which require future work to understand specific energy characteristics. However, we believe that the results using the M100 can provide important information to researchers and

industry professionals working with UAVs for last-mile deliveries for very small packages such as medical deliveries, critical parts, or other time-sensitive payloads. Here, we help stakeholders and researchers understand the energy use of uncrewed aerial package-delivery drones. We provide an energy model based on extensive empirical data from 188 flights of a quadcopter drone M100, from which we developed a high-resolution dataset of package-delivery-drone energy use.³¹ In addition, we develop an algorithm that automatically identifies the flight regime across takeoff, cruise, and landing. We show the impact of the cruise speed and payload mass on the drone's range and provide energy-use coefficients. We use our energy model to compare the energy consumption and GHG emissions of the drone with delivery trucks, delivery vans, and electric cargo bicycles on a distance and package basis. We show how the drone's emissions differ regionally in the US according to the electricity mix. We perform a sensitivity analysis on the energy consumption on a distance basis (MJ/km) and show the minimum drone energy consumption required to match different vehicles, which can help inform drone designers for future efficient UAVs. Finally, we show the delivery intensity required to match the GHG emissions of the drone for each region of the US.

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Combining Equation2 and Equation1,

$$E = \left(b_1 \frac{m^{1.5}}{\sqrt{\rho}} + b_0\right) t, \tag{Equation3}$$

where b_1 includes the constants A and g. The estimated coefficients and their standard errors are shown in Table 1.

Table 1. <u>Model coefficient</u> ± bootstrap standard error			
Coef.	Takeoff	Cruise	Landing
b ₁	80.4± 2.6	68.9± 2.0	71.5± 1.7
b ₀	13.8± 18.9	16.8± 15.0	-24.3± 12.5

Coef.	Takeoff	Cruise	Landing
R^2	0.84	0.85	0.90

The coefficients shown in Table 1 were obtained by performing a linear regression between P_i and the average power observed throughout each of the 120 flights. The results were then applied to the remaining flights, and the absolute relative error was 2.1% on average, proving the accuracy of the energy model in terms of estimation of energy consumption.

With the energy model validated, we estimated the energy consumption of a package delivered by a small quadcopter drone. Our energy model suggests that within the speed range tested, the average power consumption of a quadcopter does not vary considerably with the speed during cruise, which has also been observed with our dataset (Figure 1). The speed does affect total flight time, and hence for the drone and range of speeds we tested, total

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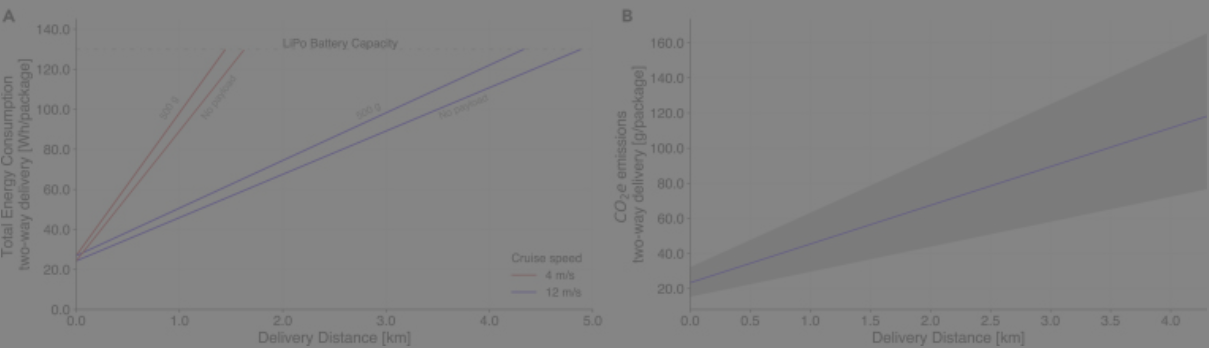
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Figure2. Small drone total energy consumption and greenhouse gas emissions by distance

(A) Total energy consumption by distance of delivery varying payload mass and cruise speed. Total trip energy is higher at lower speeds for a fixed distance.

(B) CO₂e emissions of delivering a payload of 0.5 kg at a cruise speed of 12 m/s, the fastest speed evaluated, based on the delivery distance. The uncertainty area corresponds to US regional grid emissions factors. The total energy and CO₂e correspond to takeoff, cruise from the origin to destination, and landing loaded and takeoff, cruise from destination to origin, and landing empty. As an energy limitation, the nominal capacity of an LiPO TB48D battery is 130 Wh. Results shown here have an altitude during cruise of 100 m, takeoff speed of 2.5 m/s, and landing speed of 2 m/s.

We then calculated the GHG emissions per package delivered based on the US regional non-baseload electricity

GHG emissions from the US Environmental Protection Agency (EPA) and the US regional non-baseload electricity

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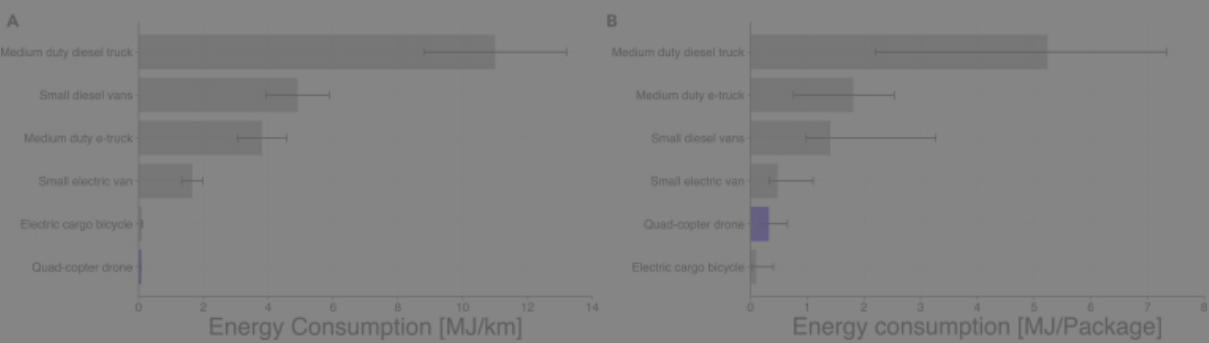
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Comparison between different transportation modes

We compared the energy consumption of quadcopter drones against [diesel](#) and electric medium-duty trucks and small vans and electric cargo bicycles.

The total energy consumption per distance of small quadcopter drones is among the lowest across [transportation modes](#), as the vehicle is small, light, and has lower [payload capacity](#) ([Figure 4A](#)). [Figure 4B](#) shows the energy consumption per package of drone-equivalent deliveries, i.e., assuming that all packages delivered by the other modes are within the payload and space capacity of a small quadcopter drone. On an energy consumption per package basis, small quadcopter drones and electric cargo bicycles are among the most energy-efficient modes for small package delivery. The number of stops per kilometer and the number of packages delivered per stop varies

according to the transportation mode and delivery density (highly dense areas are more likely to have more stops and packages delivered per kilometer).



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Figure 6. GHG emissions per distance and per package delivered for different transportation modes

Error bars (A and B) represent uncertainties due to variations on fuel carbon intensity, battery life cycle emissions, and number of packages delivered per distance.

We compared our results with values provided by the United Parcel Service (UPS). In 2019, UPS reported the energy intensity for US Domestic Package operations was 28MJ/package, from which ground vehicles represented approximately 9.5MJ/package or 34% (airline fuel, facility heating fuel, and indirect energy correspond to 60%, 3%, and 3%, respectively), with GHG emissions (CO₂e) intensity of 1 kg/package.³⁵ It is important to note that these values encompass the entire ground fleet rather than only last-mile delivery.

Finally, assuming a base case where the drone delivers an average of 0.25 packages per km (one-way delivery distance of 2 km) and consumes 0.08MJ/km, we calculated the minimum number of packages per km required by

each vehicle to match the drone’s energy consumption and CO₂e per package. Table2 shows that a medium-duty diesel truck would require approximately 34 packages per km to meet the drone’s performance, which would correspond to having 200 packages delivered in a route of less than 6km. On the other hand, a small electric van would require a delivery intensity of approximately five packages per km, or a 39-km route to deliver 200 packages, which could potentially be achieved in dense urban centers.

Table2. Delivery density of each mode to match the drone’s energy consumption per package, and the minimum drone energy consumption to match the vehicle’s energy consumption per package

Vehicle	Delivery density required to match drone energy consumption (package/km) (multiplier from base case)	Minimum energy consumption required for the drone (MJ/km)
Medium-duty diesel	33.8 (16.1×)	1.31

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per package in the central Midwest. As the electricity grid gets cleaner over time, the carbon intensity of delivery with electricity-powered vehicles, whether drone, cargo bicycle, van, or truck, will continue to improve.

Our energy model has simple and accurate coefficients that can provide stakeholders and researchers with a drone-energy-consumption estimation for speeds below 12m/s for similar small drones. However, at greater speeds or using drones with more surface area or mass, a more comprehensive energy-profile method could provide more accurate predictions.

The energy consumption of a very small, commercially available quadcopter drone with payload of 0.5kg is comparable to the most energy-efficient modes of last-mile delivery when the total mass of delivery is not the main feature considered. For example, in delivery situations where small and light items with high added value, such as medical deliveries, critical packages, and small electronics, very small drones might become a competitive

tool to reduce transportation emissions in large urban centers.³⁶ In these scenarios, we found that drones can reduce the energy consumption by 94% and 31% and GHG emissions by 84% and 29% per package delivered by replacing diesel trucks and electric vans, respectively. We also found that electric cargo bicycles had similar or lower GHGs per package than drones. We also found that the delivery intensity, i.e., the number of packages delivered per km, and the fuel carbon intensity are the main factors contributing to the drone's comparative energy and environmental performance.

Adding empirical tests to assess the impacts of varying the area of the propeller, the drone geometry and delivery intensity that were not included in this study could provide an important contribution to this field in future work. It is also important to note that the drone used to collect the data was not optimized to minimize energy consumption, which could further improve its efficiency. The drone used was designed for a very small package, and future studies should include heavier payloads. Given the potential for improved energy productivity of delivery and reduced GHGs per package, very small package delivery by drone and electric cargo bicycles can be an important tool to reduce transportation emissions in large urban centers.

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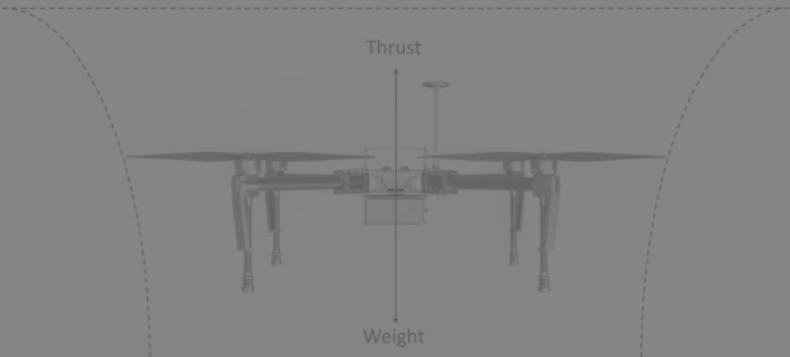
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For a better understanding of the energy-consumption profile of each flight, we created an algorithm to automatically divide the data into three different flight regimes: takeoff, cruise, and landing ([FiguresS3–S6](#)). The data available in Rodrigues et al.³⁷ were processed using this algorithm before the analysis described in the next sections.

First-principles analysis

The energy required to power a UAV can be estimated using a first-principles analysis based on helicopter aerodynamics.³⁹ First, we defined the working coordinate frames for a quadcopter drone ([FiguresS7–S9](#)). Then, we assessed the power required to maintain the drone at a steady hover condition. Finally, we expanded the power analysis to include other power demands.

The main power demand of a drone is in the form of P_i . The P_i represents the power required to overcome the force of gravity in order to keep the aircraft in the air, and it can vary according to the flight maneuver.³⁹ The most basic way to estimate P_i is considering a hover condition without wind (Figure6).



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Our energy model inquires how effectively P_i can be used as an estimator for the energy consumed during a package-delivery flight. In such a case, the average power ($\bar{P}$) throughout the flight is modeled as a linear regression of the P_i

$$\bar{P} = b_1 P_i + b_0, \tag{Equation7}$$

where b_1 and b_0 are the slope and intercept of the linear regression, respectively.

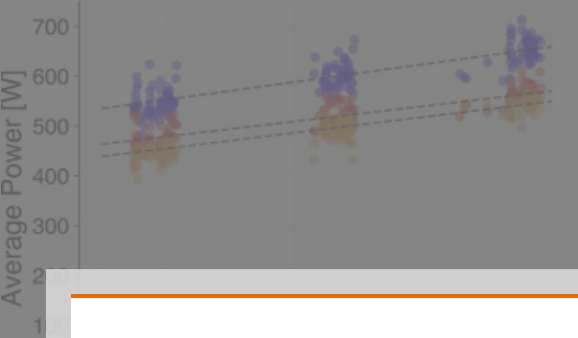
We can expand Equation7 to account for the sum of the three flight regimes (Figure7) and combine the area of the propeller (A) and gravity acceleration (g) from Equation2 with coefficient b_1 . Thus, the total energy consumption (

E) is estimated as

$$E = \sum_{r \in \mathcal{R}, l \in \mathcal{L}} \left(b_1^{(r,l)} \frac{m^{1.5(l)}}{\sqrt{p}} + b_0^{(r,l)} \right) t^{(r,l)}$$

(Equation8)

for $\mathcal{R} = \{takeoff, cruise, landing\}$ and $\mathcal{L} = \{loaded, unloaded\}$.



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tuning only, we compared the absolute relative error (ARE; [Equation9](#)) instead of the quadratic error. We fixed a regime and a set of hyperparameter values to be tuned then varied the following hyperparameters: learning rate (0.01, 0.05, 0.1), maximum depth of each weak learner (3, 5, 8), and minimum loss reduction required to make another partition (0, 1, 5). Then, we ran XGBoost on the training data using squared loss as the objective function. Our response variable and features were the same used in the first-principles energy model. We used the stochastic approach by subsampling 80% of the data at each tree fitting. Our algorithm used up to 1,000 trees; however, if there was no significant increase in performance after 50 trees, the algorithm stopped. Then, we plotted the number of trees (up to 1,000) versus training error and estimated test error via 5-fold cross validation.

After tuning, the model was trained with the optimal hyperparameters on the entire training set, and AREs were computed for the flights on the test set. We selected the hyperparameter combination to be used for each regime

(number of trees, learning rate, maximum depth, minimum loss reduction) by choosing a low test error, low generalization error, and low standard errors. The selected hyperparameter values are in TableS1.

Estimation of the standard errors for the coefficients

To obtain standard errors of the estimated coefficients, we used a nonparametric bootstrap approach.⁴¹ 1,000 bootstrap replications were used to resample with replacement the 120 training flights, and the energy models for the three flight regimes were refitted for each bootstrap sample. At the end, the standard errors (SEs) of the coefficients were obtained from their sampling distribution.

Model predictive power

To evaluate the model's predictive power, the regime-specific fitted models were then applied to the testing flights of the test set and were compared by ARE computed at flight resolution. That is, for each flight from the test set, we

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$$E = \left(b_1 \frac{m^{1.5}}{\sqrt{\rho}} + b_0\right) \frac{h}{V},$$

(Equation 12)

where h is the cruise altitude and V is the average speed during takeoff and landing.

For instance, a small quadcopter operating at V_{cr} = 12m/s, payload= 500 g, h= 100 m, takeoff average speed (V_{tk})= 2.5m/s, and landing average speed V_{ld} = 2.0m/s has a range of approximately 4km (2km of one-way delivery range), consuming approximately 74.5 Wh (per round-trip delivery).

Therefore, a quadcopter drone flying under these conditions would consume approximately 0.067MJ/km, not considering charging and transmission losses. The energy consumption during takeoff and landing for this trip corresponds to approximately 36% of the total energy consumption (26.8 Wh per trip). This share of energy could be reduced by 95% in a 5-m takeoff (to 1.34 Wh per trip), which could be achieved, for instance, if the drone would

depart from the top of a building, provided sufficient space for safe operation. This would reduce total trip energy by 34% (74.5–48.95 Wh) or increase the range from 2 to 3km (4–6km two-way distance), consuming the same amount of energy. Further analysis of system-wide energy benefits of building-to-building drone travel is an important topic for future research.

Transport mode comparison

We develop a model comparing the small quadcopter drone with different transportation modes in terms of energy consumption and CO₂e emissions and validate it against top-down [sustainability](#) reports from UPS in 2019. The energy consumption of a medium-duty diesel truck is considered as 11MJ/km⁴³. Whereas a medium-duty electric truck has an energy consumption of 1.4kWh/mile,⁴² or 3.13MJ/km, diesel vans operate at 18.4 MPG on average,⁴³ or 4.9MJ/km (conversion factors: one gallon of diesel= 137,381 Btu,⁴⁴ 1MJ= 947.817 Btu, one mile= 1.60934km). On the other hand, electric vans operate with an energy consumption of 0.38kWh/km⁴⁵ or 1.36MJ/km. Finally, an electric bicycle has an energy consumption of 0.08kWh/km⁴⁶ or 0.3MJ/km.

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The electricity CO₂e emissions were considered to be the 2020 American non-baseload average of 177 g/MJ (638 g/kWh), with the lower bound being 111 g/MJ (399 g/kWh) from New England and the [upper limit](#) being 250 g/MJ (900 g/kWh), reflecting non-baseload emissions from the central Midwest.³² CO₂e emissions for diesel combustion was considered as 1.61×10⁻⁴ lb/Btu,⁵¹ or 69.35 g/MJ. Upstream GHG emissions for average diesel and [electricity generation](#) are 15 g/MJ and 22 g/MJ⁵³, respectively. The battery life cycle emissions for the drone (assumed to be similar to Li-iron phosphate) were calculated as 0.76 (base case), 0.23 (low case), and 1.52 g/km (high case). Similarly, the electric cargo bicycle has battery life cycle emissions of 1.3 g/km, considering a Li-ion NMC811

battery. For the electric van and electric medium-duty truck, we assumed a battery of Li-ion NMC811, resulting in 14.1 g/km for the van and 24.5 g/km for the truck.⁵² Energy and emissions associated with system changes beyond the vehicle (such as in warehousing²⁰) are not included and are important areas for future work.

The energy consumption per package (E_{pack}) was calculated as

$$E_{pack} = \frac{E_{dist}}{S_{freq} \cdot P_{freq}},$$

(Equation 13)

where E_{dist} is the energy consumption per distance unit, S_{freq} is the number of stops to deliver packages per distance unit, and P_{freq} is number of packages delivered per stop on average.

Similarly, the GHG emissions per package (GHG_{pack}) is calculated as

$$GHG_{pack} = \frac{E_{dist} \cdot GHG_{energy}}{S_{freq} \cdot P_{freq}}$$

(Equation 14)

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Declaration of interests

The authors declare no competing interests. C.S. has a patent for a system, method, and computer program product for transporting an unmanned vehicle, US patent 11300978B1, 2022. The patent's methods were not used this study.

Inclusion and diversity

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in science. One or more of the authors of this paper self-identifies as a member of the LGBTQ+ community.

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Data and code availability

All data and code are available at <https://www.tandfonline.com/doi/full/10.1080/10447318.2024.2384452> and all modeling codes are available at <https://github.com/10447318.2024.2384452>.

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